Chapter 3 - Conditional Statements

What are Conditional Statements?

Conditional statements let a program make decisions. The program checks a condition, which is an expression that evaluates to True or False, and runs different code depending on the result. Python provides the if, elif and else keywords for this purpose.

```
marks = 72
if marks >= 80:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Grade C")
```

How if, elif and else Work

- if: runs its block when the condition is True.
- elif (else if): checked only when the previous conditions were False.
- else: runs when none of the conditions were True; it is optional.
- Only the first block whose condition is True is executed.

Comparison and Logical Operators

Conditions are built with comparison operators: ==, !=, >, <, >= and <=. They can be combined with the logical operators and, or and not. Remember that = assigns a value while == compares two values.

```
age = 19
has_id = True
if age >= 18 and has_id:
    print("You may enter")
if not has_id:
    print("ID required")
```

Nested Conditions and the Ternary Expression

An if statement can be placed inside another if statement. Python also offers a one-line conditional expression: value_if_true if condition else value_if_false.

```
status = "Pass" if marks >= 40 else "Fail"
```

Truthy and Falsy Values

Every Python value can be used as a condition. Falsy values are False, None, 0, 0.0, empty strings, empty lists, empty tuples and empty dictionaries. Everything else is truthy. For example if my_list: is True only when the list has items.

The match Statement

Since Python 3.10 the match statement provides pattern matching, similar to switch in other languages.

```
day = 3
match day:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
    case _:
        print("Another day")
```